Note 16: Counting Sort

Instructions

Our objective is to study and define the *counting sort* and its variations.

Background

Any sorting algorithm that sorts by comparing the individual elements of a collection is called a *comparison sort*. Every comparison sort requires $\Omega(n \lg n)$ comparisons in the worst case. Better running times are possible when the input is constrained, as in the counting sort. If the collection consists only of integers in the range $[0, k-1]$, the counting sort has a running time of $\Theta(n+k)$, where n is the length of the collection and k is the number of possible values. Its algorithm for sorting an array A of integers in $[0, k-1]$ into an output array B is:

```
CountingSort(A, B, k)
  //Initialize every element of the counter array C to zero
  i ← 0
  while i < k, do
    C[i] ← 0
    i ← i + 1
  //Store the count of each distinct value of A in C
  i ← 0
  while i < A.length, do
    C[A[i]] ← C[A[i]] + 1
    i ← i + 1
  //Adjust C so that C[i] is the number of values less than or equal to i
  i ← 1
  while i < k, do
    C[i] ← C[i] + C[i - 1]
    i ← i + 1
  //Place each element of A into its sorted position in B, from last to first
  i ← A.length
  while i > 0, do
    C[A[i - 1]] ← C[A[i - 1]] - 1
    B[C[A[i - 1]]] ← A[i - 1]
    i ← i - 1
```

The counting sort sorts without comparing individual elements, and it is *stable*, which means that elements with equal values keep their original relative order. For example, if a_1 and a_2 are equal values with a_1 appearing first, and $a < b < c < d$, then $A = [c, a_1, b, d, a_2]$ becomes $A = [a_1, a_2, b, c, d]$ after the sort.

Tracing

To verify that the algorithm is correct, trace by hand the contents of the arrays A, B, and C after each loop terminates, for each of the following initial arrays A. Use $k = 10$.

1. $A = [8, 9, 2, 7, 7, 1, 8, 3, 5, 4, 7, 9, 3, 6, 6, 9, 5]$
2. $A = [9, 8, 8, 0, 3, 9, 8, 7, 7, 8]$
3. $A = [5, 7, 9, 7, 5]$
4. $A = [5, 8, 9, 4, 5, 8, 9, 2, 0, 5, 1, 8, 1, 1, 8]$
5. $A = [0, 7, 8, 3, 3, 3, 9, 2, 0, 9]$